Feasibility of Ion-Thruster-Only Maneuvering for a 6U CubeSat in Very Low Earth Orbit

Jonathan Stahl*, David Franklin II*, Sai Vidyud Senthil Nathan*, Aiden Reckamp*, Thanadis Charoenrujijin*, and Connor Lashley*
The Ohio State University, Columbus, Ohio 43210

This paper evaluates whether ion thrusters alone can support both agile attitude maneuvers and sustained operation for a 12 kg**, 6U CubeSat in very low Earth orbit (VLEO). The study combines a minimax attitude-slew model, target-of-opportunity tracking simulations, and a free-molecular aerodynamic compensation survey. A representative actuator-force budget of** 145 mN **is applied through moment arms of** [0.3, 0.3, 0.2] **m. The instantaneous maneuver results are favorable: a** 180° **rest-to-rest slew completes in** 6.13 s **on the worst principal axis and** 3.97 s **on the optimal axis, while illustrative** 200 km **tracking and slew-to-track cases require only** 0.685 mN **and** 0.617 mN **peak actuator force, respectively. An aerodynamic-aware match-time scan places the slew-to-track handoff at** 193 s **and reveals that the required actuator force is dominated by aerodynamic compensation rather than by the maneuver itself. However, sustained VLEO operation is endurance-limited rather than peak-force-limited. For a** 1.5 kg **usable propellant allocation, the nadir-pointing mission life at** 170 km **is only 6.74 days at** $I_{sp} = 760$ s **and 26.62 days at** $I_{sp} = 3{,}000$ s**; even at** 200 km**, the same cases yield only 18.70 and 73.82 days. An advanced** $I_{sp} = 19{,}300$ s **case extends life substantially but is not representative of current CubeSat-class ion propulsion. The results demonstrate that ion thrusters can satisfy the instantaneous force demand for VLEO maneuvers, but ion-thruster-only architectures are not well suited to sustained VLEO missions unless paired with higher propellant mass, substantially higher specific impulse, or hybrid control architectures.**

I. Nomenclature

$\boldsymbol{\beta}$	=	quaternion attitude vector
$\boldsymbol{\omega}$	=	angular velocity vector, rad/s
$\boldsymbol{H}_C$	=	angular momentum about center of mass
$F_{\max}$	=	actuator force budget per controlled axis
$\bar{F}_{\text{tot}}$	=	orbit-averaged total thruster force
g_0	=	standard gravitational acceleration, 9.806,65 m s^{-2}
$\mathbf{I}_C$	=	spacecraft body-frame inertia matrix
I_{sp}	=	specific impulse
L_i	=	effective moment arm for body axis i
m_{prop}	=	usable propellant mass
$\dot{m}$	=	propellant mass flow rate
$T_{\min}$	=	minimum rest-to-rest slew time
t_{life}	=	propellant-limited mission life
θ	=	angular separation between initial and target attitudes
τ_i	=	body-axis control torque component
γ	=	minimax body-axis force demand, $\max_i \lvert\tau_i\rvert / L_i$
TOO	=	target of opportunity
VLEO	=	very low Earth orbit

*Student, Department of Mechanical and Aerospace Engineering, The Ohio State University.

II. Introduction

Very low Earth orbit (VLEO) offers compelling advantages for Earth observation missions, including reduced launch costs for heavier payloads, enhanced geospatial imaging resolution, and improved signal-to-noise ratios for ground-directed sensors. These characteristics make VLEO particularly attractive for rapid-response applications such as wildfire detection, natural disaster coordination, and atmospheric and space-weather monitoring. However, operating below approximately 300 km introduces severe environmental challenges: residual atmospheric density generates substantial aerodynamic drag that continuously depletes orbital energy, while the atomic-oxygen flux accelerates material degradation of exposed spacecraft surfaces.

JAXA's TSUBAME (SLATS) technology demonstrator underscores both the promise and the difficulty of the regime. TSUBAME reached a record low sustained orbital altitude of 167.4 km, and JAXA reported that the 200 km to 300 km environment can impose atmospheric and atomic-oxygen exposure roughly three orders of magnitude more severe than that encountered in conventional Earth-observation orbits [1, 2].

Ion thrusters are attractive for VLEO because of their high specific impulse, and representative hardware already spans a wide performance range. NASA's NEXT-C operates over approximately 25 mN to 235 mN [3]. QinetiQ's T6 has published operating points up to approximately 145 mN [4]. At the lower end, electrospray devices can operate near $I_{sp} = 760\,\mathrm{s}$ [5]; conventional low-power ion engines have historically occupied the 2,000 s to 3,000 s range [6]; and ESA's DS4G concept demonstrates a substantially higher upper bound near 19,300 s [7]. Despite their high fuel efficiency, ion thrusters produce low thrust relative to chemical systems, and their substantial electrical power requirements complicate accommodation on compact platforms [8, 9].

The central engineering question is not whether ion thrusters can generate thrust, but whether they alone can provide sufficient instantaneous control authority for agile VLEO maneuvers while simultaneously preserving useful mission life. Peak force determines whether a given maneuver is feasible at all, whereas cumulative propellant consumption determines whether the mission can remain in the useful VLEO altitude band long enough to fulfill its objectives.

This paper evaluates that trade for a 12 kg 6U CubeSat through four linked studies:

1) a force-limited analytic and numerical rest-to-rest slew sweep,
2) an altitude sweep for nadir-pass target-of-opportunity (TOO) tracking with aerodynamic compensation,
3) deterministic 200 km TOO tracking and slew-to-track demonstrations, and
4) a propellant-limited endurance sweep for continuous aerodynamic compensation under nadir-pointing operation.

The central finding is that ion thrusters can satisfy the instantaneous force demand for the modeled VLEO maneuvers, but an ion-thruster-only architecture becomes life-limited at practically interesting VLEO altitudes, suggesting that hybrid control strategies or substantially higher-performance propulsion are necessary for operationally viable missions.

III. Background

A. Ion propulsion in the context of VLEO attitude control

Ion thrusters have traditionally been employed for deep-space propulsion and long-duration orbital station-keeping, applications in which their exceptionally high specific impulse translates directly into reduced propellant mass [8]. However, their characteristically low thrust presents a distinct challenge for VLEO attitude-control applications, where rapid reorientation toward ground targets demands agile torque authority. Additionally, small satellites operating in VLEO are subject to stringent mass and volume constraints, and the substantial electrical power required by ion systems, which typically necessitate large solar-array areas, complicates integration on compact platforms [9]. This study evaluates whether the modest thrust levels characteristic of ion propulsion can nevertheless provide adequate torque for the required slew and tracking maneuvers.

B. Attitude kinematics

Multiple parameterizations exist for representing the rotational state of a rigid body in three-dimensional space, including Euler angles, direction cosine matrices, and quaternions [10]. Euler angles are intuitive but suffer from singularities at certain orientations regardless of the rotation sequence chosen. Direction cosine matrices avoid singularities through redundancy (nine elements for three rotational degrees of freedom) at the cost of increased computational complexity. Quaternions offer a favorable compromise: four parameters provide a singularity-free, minimal-redundancy representation of attitude.

A quaternion $\boldsymbol{\beta} = [q_0,\ q_1,\ q_2,\ q_3]^T$ encodes the orientation of the body frame relative to a reference frame using the scalar-first convention. The kinematic differential equation relating the quaternion rate to the body angular velocity $\boldsymbol{\omega}$ is

$$\dot{\boldsymbol{\beta}} = \frac{1}{2}\, B[\boldsymbol{\beta}] \begin{pmatrix} 0 \\ \boldsymbol{\omega} \end{pmatrix}, \tag{1}$$

where $B[\boldsymbol{\beta}]$ is a 4×4 matrix constructed from the quaternion elements. Because numerical integration introduces norm drift, the quaternion is renormalized after each propagation step to enforce the unit-norm constraint $\|\boldsymbol{\beta}\| = 1$.

C. Attitude kinetics

The rotational equations of motion are derived from the angular momentum principle [11]. The angular momentum of a rigid body about its center of mass C is

$$\boldsymbol{H}_C = \mathbf{I}_C\, \boldsymbol{\omega}, \tag{2}$$

where $\mathbf{I}_C$ is the inertia matrix expressed in the body frame. Applying Newton's second law for rotation and invoking the transport theorem to express the time derivative in the rotating body frame yields

$$\left(\frac{d\boldsymbol{H}_C}{dt}\right)_{\text{inertial}} = \left(\frac{d\boldsymbol{H}_C}{dt}\right)_{\text{body}} + \boldsymbol{\omega} \times \boldsymbol{H}_C. \tag{3}$$

Equating the inertial rate of change of angular momentum to the sum of external torques $\sum \boldsymbol{L}_C$ and substituting Eq. (2) produces Euler's rotational equation:

$$\sum \boldsymbol{L}_C = \mathbf{I}_C\, \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \left(\mathbf{I}_C\, \boldsymbol{\omega}\right). \tag{4}$$

Rearranging to isolate the angular acceleration provides the state-space form used for numerical integration:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}_C^{-1}\left(\sum \boldsymbol{L}_C - \boldsymbol{\omega} \times \left(\mathbf{I}_C\, \boldsymbol{\omega}\right)\right). \tag{5}$$

Together, Eqs. (1) and (5) form the seven-state attitude propagation system $[\boldsymbol{\beta};\, \boldsymbol{\omega}]$ that underlies the simulations presented in this paper.

D. Aerodynamic environment in VLEO

In VLEO, aerodynamic drag constitutes the dominant environmental disturbance acting on a spacecraft. Although the atmosphere is highly rarefied at altitudes below 300 km, the residual gas density, combined with orbital velocities exceeding $7.5\,\mathrm{km\,s^{-1}}$, generates substantial drag forces and torques that continuously deplete orbital energy and perturb the spacecraft attitude. The magnitude of these loads depends on atmospheric density, spacecraft velocity, the exposed cross-sectional area, and the gas–surface interaction model. Because VLEO atmospheric density varies significantly with solar activity, geomagnetic conditions, and diurnal effects, accurate drag prediction requires dynamic atmospheric modeling rather than static density assumptions.

At the altitudes considered in this study, the mean free path of atmospheric molecules greatly exceeds the characteristic dimension of the spacecraft, placing the flow firmly in the free-molecular regime (Knudsen number $\gg 1$). Under these conditions, the Navier–Stokes continuum framework is inapplicable; instead, gas–surface interaction models such as the Sentman or Cook formulations are used to compute the aerodynamic force and torque on each exposed panel individually.

Other environmental disturbances, including gravity-gradient torques, residual magnetic dipole interactions, and solar radiation pressure, also influence attitude dynamics, but are typically at least an order of magnitude smaller than aerodynamic torques at VLEO altitudes and are not modeled explicitly in the present analysis.

IV. Methodology

Figure 1 provides an overview of the analysis workflow used in this study.

A. Spacecraft and actuator model

The reference vehicle is a $10 \times 20 \times 30$ cm, 12 kg 6U CubeSat. The principal inertia matrix is

$$\mathbf{I}_C = \mathrm{diag}(0.13,\ 0.10,\ 0.05)\ \mathrm{kg\,m^2}. \tag{6}$$

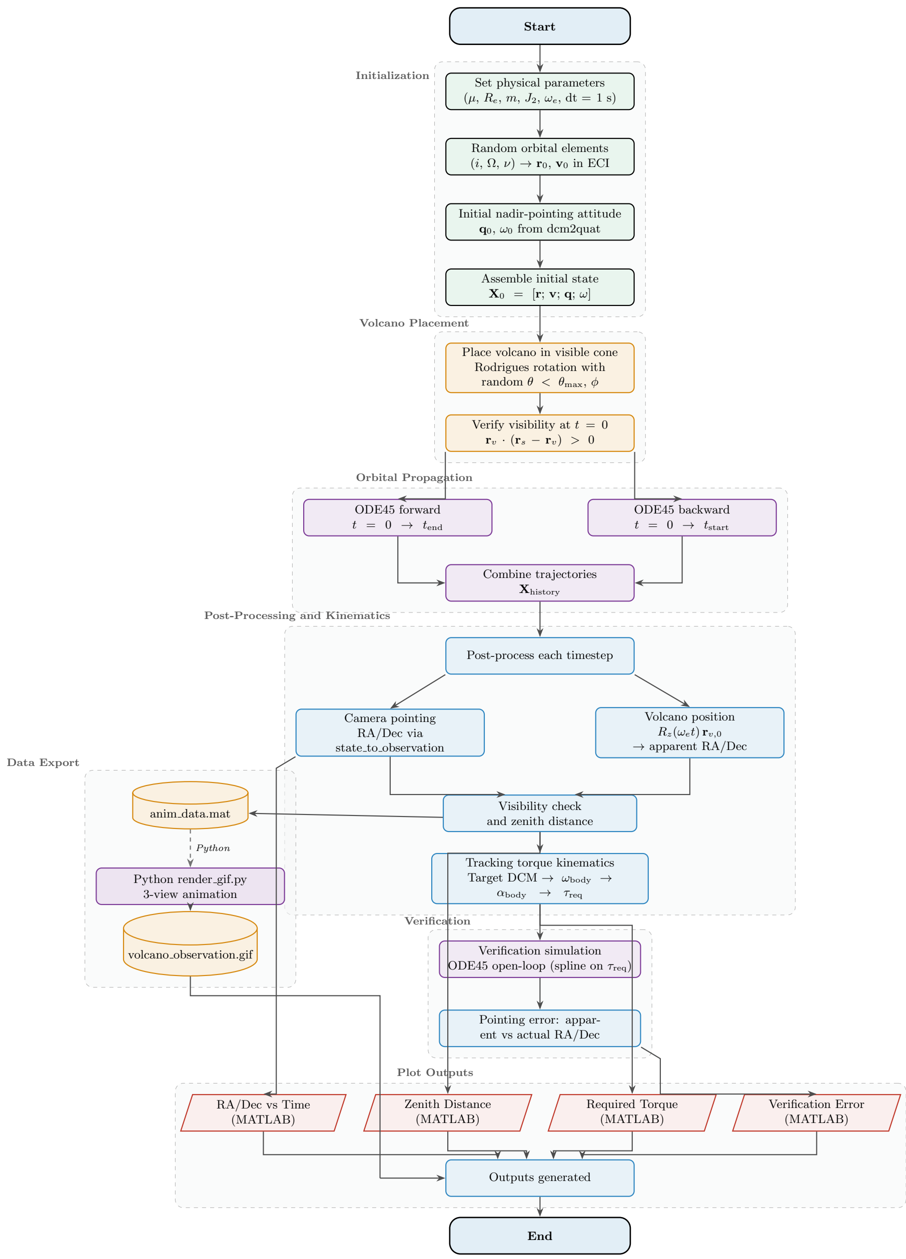

Fig. 1 Overview of the analysis workflow. (Placeholder; to be replaced with final flowchart.)

The actuation geometry is represented by effective moment arms $\mathbf{L} = [0.3,\ 0.3,\ 0.2]^T$ m, so that the body-axis force demand for a given torque vector is

$$\gamma = \max_i \frac{|\tau_i|}{L_i}. \tag{7}$$

Throughout this paper, the baseline actuator-force budget is $F_{\max} = 145$ mN, chosen to be consistent with a high-end demonstrated ion-engine operating point [4].

For rest-to-rest principal-axis maneuvers, the analytic minimum slew time is

$$T_{\min,k} = 2\sqrt{\frac{\theta\, I_k}{L_k\, F_{\max}}}, \tag{8}$$

where θ is the rotation angle about principal axis k. For finite-time slew-to-track transitions, a minimax boundary-value solver computes the torque history that minimizes the peak body-axis force demand, with the resulting control torques converted to equivalent actuator force through Eq. (7).

B. Scenario set

Four scenario families are evaluated.

Rest-to-rest sweep. The minimum slew duration is computed for angular separations from 0.5° to 180°, isolating the pure attitude-agility limit imposed by the force budget.

TOO altitude sweep. A nadir-pass target-of-opportunity scenario places the ground target directly below the spacecraft at closest approach and computes the maximum actuator force required to maintain tracking throughout the visible pass. The sweep spans 100 km to 200 km altitude.

Deterministic 200 km *demonstrations.* Two illustrative cases at 200 km are simulated using a fixed scenario seed for reproducibility. The first case performs continuous TOO tracking; the second starts from a nadir-pointing hold, executes a finite-time minimax slew, and hands off into the tracking attitude and body rate. The seeded geometry is comparatively benign and should be interpreted as illustrative rather than worst-case.

Nadir-pointing endurance sweep. Propellant-limited mission life is evaluated from 150 km to 250 km. Each altitude point is averaged over three circular orbits with 120 samples per orbit. The survey assumes 1.5 kg usable propellant and evaluates three representative specific-impulse values: $I_{sp} = 760$ s (electrospray-class lower bound [5]), $I_{sp} = 3{,}000$ s (conventional ion-thruster class [6]), and $I_{sp} = 19{,}300$ s (advanced DS4G-class upper bound [7]).

C. Slew maneuver design

D. Target tracking control

E. Aerodynamic compensation and endurance model

Aerodynamic loads are evaluated in the free-molecular regime using the Cook gas–surface interaction model. The endurance survey allocates both the aerodynamic force and the aerodynamic torque to an eight-thruster corner layout, then uses the orbit-averaged total scalar thruster force $\bar{F}_{\text{tot}}$ to estimate propellant consumption and mission life:

$$\dot{m} = \frac{\bar{F}_{\text{tot}}}{I_{sp}\, g_0}, \qquad t_{\text{life}} = \frac{m_{\text{prop}}}{\dot{m}}. \tag{9}$$

This formulation is intentionally conservative: it does not consider whether a hybrid attitude-control architecture could reduce propellant consumption, but instead evaluates whether ion thrusters alone can both maneuver and sustain the vehicle in VLEO.

Table 2 Representative quantitative results from the VLEO ion-thruster feasibility study.

Metric	Value
Worst-axis 180° rest-to-rest slew time at 145 mN	6.13 s
Optimal-axis 180° rest-to-rest slew time at 145 mN	3.97 s
Nadir-pass TOO tracking altitude where peak force crosses 145 mN	112.7 km
Illustrative 200 km TOO tracking peak force / max angular error	0.685 mN / 0.010°
Illustrative 200 km slew-to-track peak force / post-match max pointing error	0.617 mN / 0.018,3°
Mission life at 170 km for I_{sp} = [760, 3000, 19300] s	[6.74, 26.62, 171.24] days
Mission life at 200 km for I_{sp} = [760, 3000, 19300] s	[18.70, 73.82, 474.89] days

V. Results

A. Force-limited maneuver envelope

Figure 2 presents the attitude-agility result for the rest-to-rest sweep. With the 145 mN force budget, the minimum-time response is fast across the full angular-separation range. Even the worst principal axis completes a 180° slew in 6.13 s, while the optimal-axis maneuver completes in 3.97 s. The analytic and numerical solutions are in close agreement throughout, and no region of the sweep indicates that instantaneous slew authority is a limiting factor for the modeled 6U vehicle.

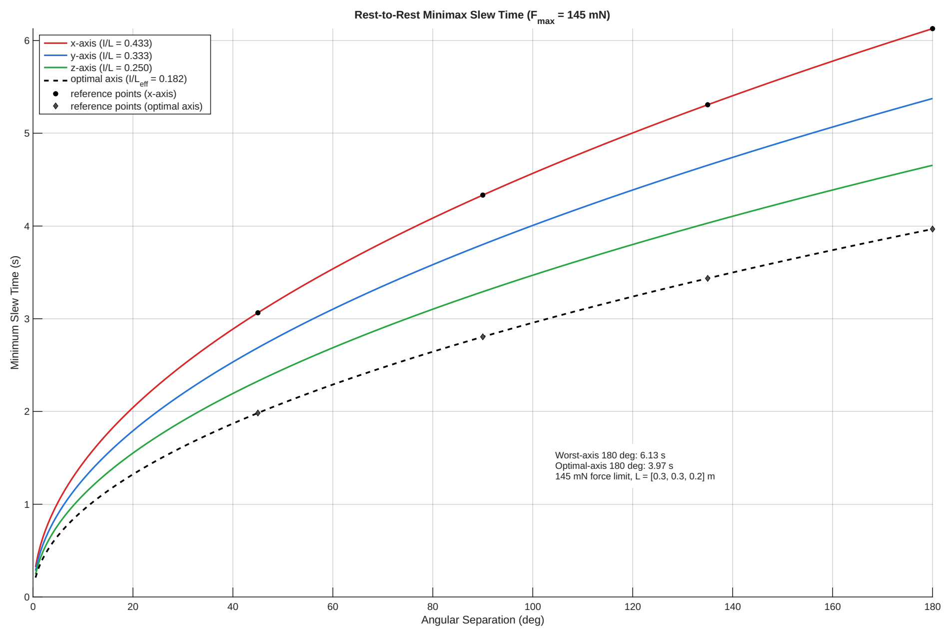

Fig. 2 Force-limited minimum slew time versus angular separation for the 12 kg **6U reference vehicle. The** 145 mN **force budget permits sub-**7 s **rest-to-rest slews even for a worst-axis** 180° **maneuver.**

This result establishes that peak-force feasibility is not the binding constraint for the modeled vehicle. Were the system peak-force-limited, the subsequent VLEO analyses would be moot. Instead, the sweep confirms that a high-end ion-thruster-class force budget is already sufficient for rapid rigid-body reorientation across the full angular range.

B. Altitude-dependent TOO tracking feasibility

Figure 3 extends the force analysis to a more operationally representative scenario: nadir-pass TOO tracking with aerodynamic compensation. The peak force requirement is strongly altitude-dependent. The 145 mN budget is exceeded at approximately 112.7 km. Above that altitude, the required force decreases rapidly: 55.2 mN at 120 km, 5.0 mN at 150 km, and only 0.94 mN at 200 km.

This threshold establishes that above the deepest VLEO regime, the tracking problem is not peak-force-limited. An ion-thruster-class actuator can sustain the modeled nadir-pass tracking geometry without saturating, even after aerodynamic disturbances are included.

C. Illustrative 200 km tracking and slew-to-track cases

While the altitude sweep addresses the envelope question, deterministic time histories provide additional insight into the maneuver structure. Figure 4 shows the seeded 200 km continuous-tracking case. The peak actuator force is 0.685 mN, corresponding to 0.47% of the 145 mN budget, and the open-loop verification error remains near 0.010°. Although this particular seed represents a comparatively favorable geometry, the result demonstrates that the force demand is far below the representative ion-thruster limit.

The more operationally demanding case is the slew-to-track handoff. A nominal match time is selected from the visibility geometry using a three-quarter visibility rule, and nearby feasible solutions are scanned to confirm the local

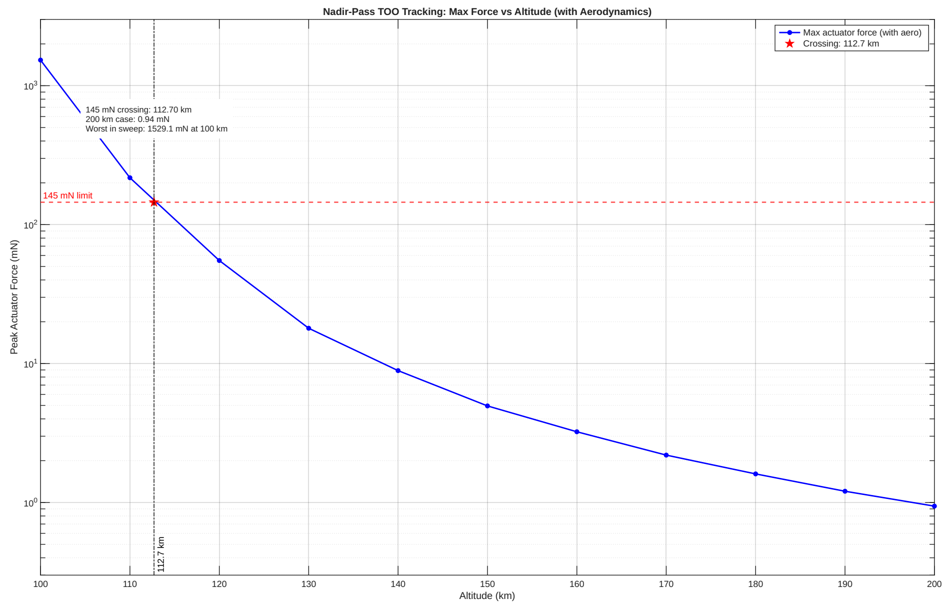

Fig. 3 Peak actuator force required for nadir-pass TOO tracking with aerodynamic compensation. The 145 mN **budget is exceeded only below approximately** 112.7 km**.**

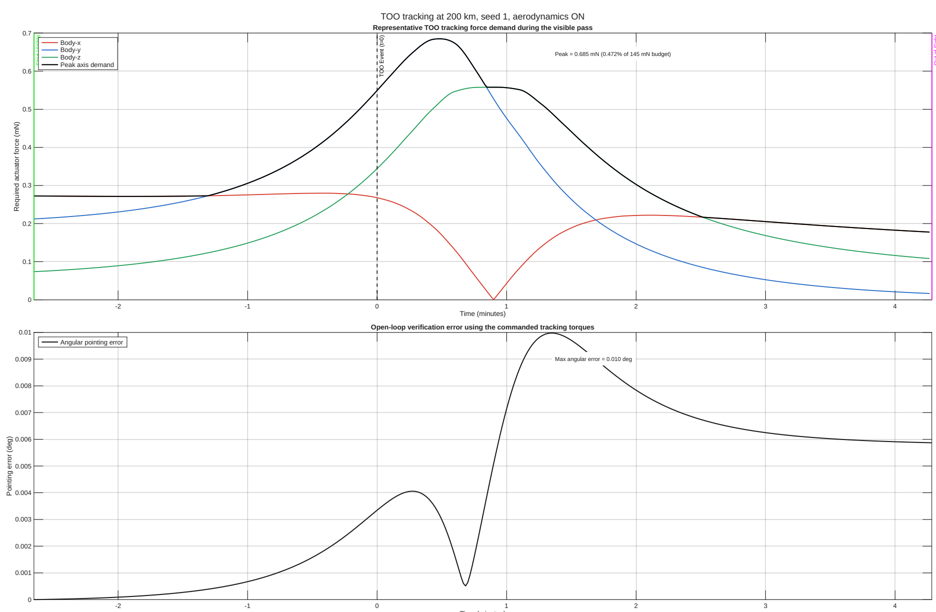

Fig. 4 Seeded 200 km **TOO continuous-tracking case. The peak actuator force remains below** 1 mN**, and the open-loop verification accuracy is approximately** $0.010°$**.**

behavior of the minimax objective. Figure 5 shows the seeded 200 km case with aerodynamic compensation included in the scan metric. The selected match occurs at 193 s (3.22 min) after event onset. The scan reveals that the full peak actuator-force equivalent is approximately 0.615 mN, while the maneuver-only component contributes approximately 0.071 mN; aerodynamic compensation therefore dominates the total force requirement.

Figure 6 shows the corresponding torque history. The match attitude error is approximately 4.91×10^{-5} deg, the post-match maximum pointing error is $0.018{,}3°$, and the full-case peak actuator force is 0.617 mN, only 0.43% of the 145 mN budget.

Collectively, the tracking and slew-to-track results reinforce the same conclusion as the analytic and altitude sweeps:

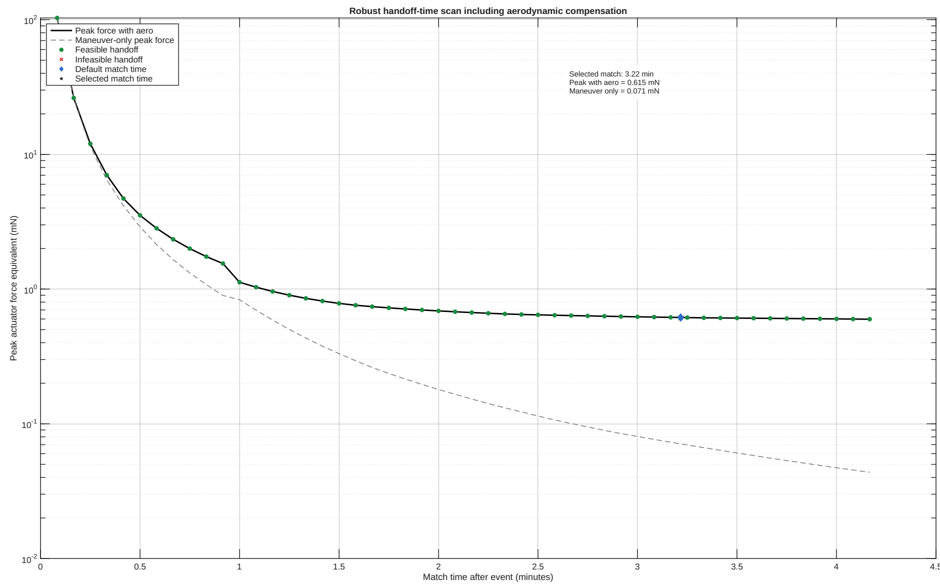

Fig. 5 Seeded 200 km **match-time scan for the TOO slew-to-track transition. The solid curve includes aerodynamic compensation; the dashed curve shows the maneuver-only component. The default and selected match times coincide for this seed.**

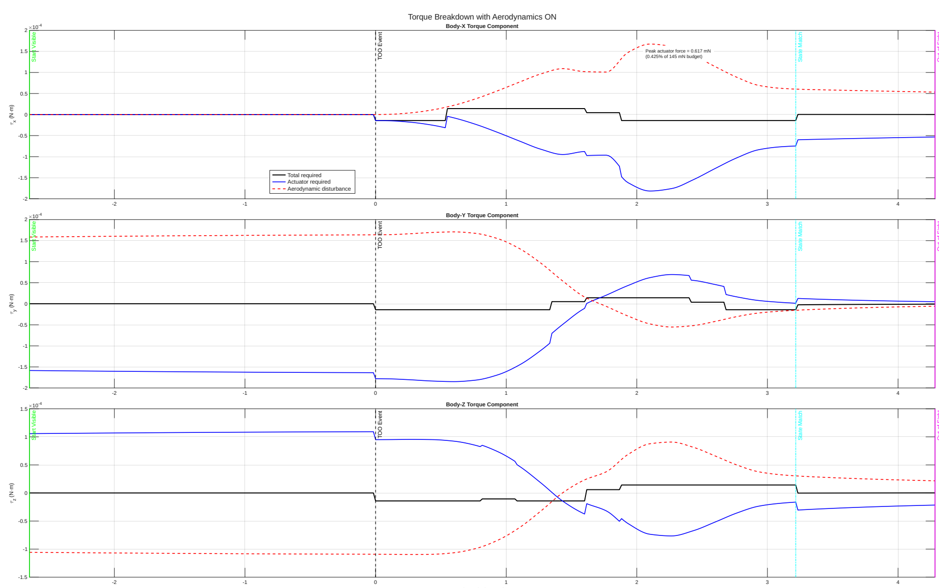

Fig. 6 Seeded 200 km **slew-to-track torque history. Aerodynamic compensation dominates the torque budget; the full case remains below a** 0.617 mN **peak actuator-force equivalent.**

instantaneous force authority is not the dominant constraint for the modeled 200 km VLEO mission.

D. Endurance limits under continuous aerodynamic compensation

The endurance analysis yields a qualitatively different picture. Figure 7 shows that the orbit-averaged total thrust required to offset aerodynamic forces and torques decreases strongly with altitude, but the resulting mission life remains short for low- to moderate-I_{sp} ion thrusters.

At 170 km, the average total thrust requirement is 19.188 mN, yielding a propellant-limited life of only 6.74 days for the I_{sp} = 760 s case and 26.62 days for the I_{sp} = 3,000 s case. At 200 km, the average total thrust decreases to

6.919 mN, but life remains only 18.70 and 73.82 days for the same two cases. The advanced I_{sp} = 19,300 s case is substantially more favorable (171.24 days at 170 km and 474.89 days at 200 km), but this performance level corresponds to an advanced high-power concept rather than a near-term CubeSat-class baseline [7].

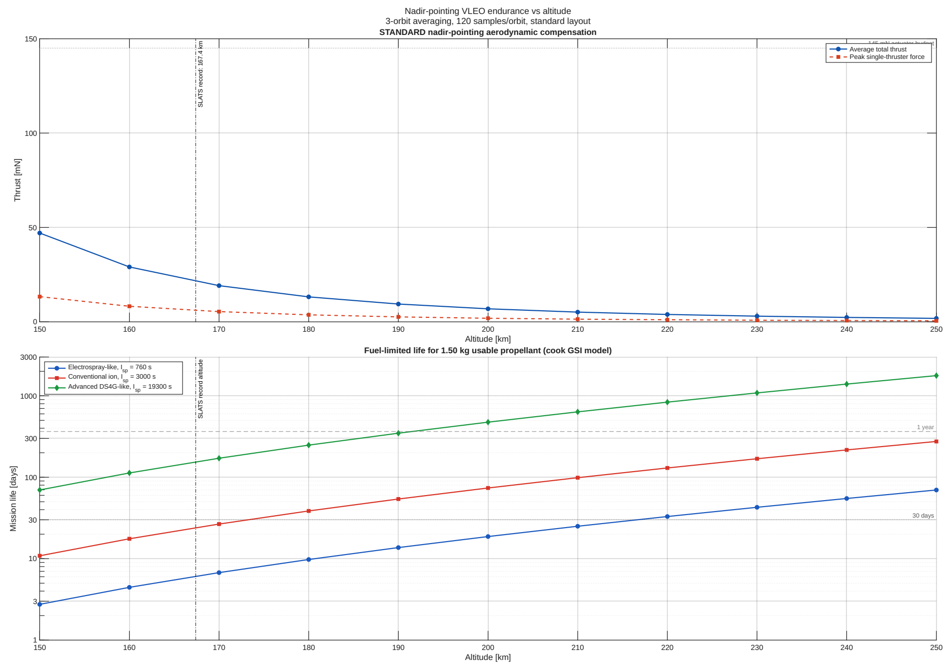

Fig. 7 Nadir-pointing VLEO endurance versus altitude for three specific-impulse cases. The force demand is modest above the deepest VLEO regime, but sustained aerodynamic compensation drives short mission life for the I_{sp} = 760 s and I_{sp} = 3,000 s cases.

This endurance result is the central finding of the study. It demonstrates that VLEO maneuvering and tracking can be feasible from a force standpoint while remaining operationally constrained by propellant-limited mission duration.

VI. Discussion

The results delineate two distinct regimes for the VLEO ion-thruster problem.

In the peak-force regime, the findings are encouraging. For the modeled 6U vehicle, the 145 mN budget provides ample margin for representative 200 km tracking and event-driven slew-to-track operations, and it remains adequate for nadir-pass TOO tracking down to approximately 112.7 km. The modeled maneuver force requirements are therefore compatible with high-end ion-thruster-class actuation.

In the endurance regime, the findings are unfavorable. The VLEO aerodynamic environment imposes continuous propellant consumption, and the resulting mission lives are short for the I_{sp} = 760 s and I_{sp} = 3,000 s cases. The governing design question is therefore not whether the ion thruster can execute a given maneuver, but how long the mission can remain in the useful VLEO altitude band while performing those maneuvers. For the configuration studied here, the answer is typically measured in days to weeks rather than in operationally meaningful durations.

These findings carry several system-level implications:

1) Ion thrusters are plausible as a source of VLEO attitude-control authority, particularly for large-angle slews and aerodynamic disturbance rejection.
2) Ion thrusters alone are a poor match for sustained low-altitude missions unless the spacecraft carries a substantially larger propellant fraction, operates at higher altitude, or employs propulsion with dramatically higher I_{sp} than conventional CubeSat-class ion systems.
3) Hybrid architectures remain the more attractive system-level solution. Reaction wheels or other momentum-storage devices can absorb fast attitude transients, while electric propulsion is reserved for drag compensation and slower momentum management.

The study also has limitations that should be acknowledged. It does not size power-processing units, solar arrays, or thermal-rejection hardware; it does not model throttling bandwidth, plume impingement, minimum impulse bit, or duty-cycle constraints; and it uses a single deterministic seeded tracking geometry for the illustrative figures. These omissions do not invalidate the central conclusions, but they are relevant if the analysis is extended from force-feasibility assessment to flight-system design. In practice, these additional constraints are likely to make ion-thruster-only architectures less attractive rather than more so.

VII. Conclusion

This paper evaluated ion-thruster-only maneuvering for a 12 kg 6U CubeSat in VLEO using analytic slew limits, target-tracking simulations, and propellant-limited endurance estimates. The instantaneous maneuver results are favorable: worst-axis 180° slews complete in 6.13 s, the illustrative 200 km tracking case requires only 0.685 mN, and the slew-to-track handoff requires 0.617 mN peak actuator force with an aerodynamic-aware scan selecting a 193 s match time. In that force-feasibility sense, VLEO maneuvering is achievable with ion-thruster-class actuation.

The mission-life result constitutes the dominant constraint. Continuous aerodynamic compensation in VLEO drives short endurance for the $I_{sp} = 760$ s and $I_{sp} = 3{,}000$ s cases, even though the corresponding peak maneuver forces remain well within budget. Ion thrusters can therefore render VLEO maneuvers feasible, but ion thrusters alone do not make sustained VLEO missions attractive for the modeled CubeSat configuration. For practical VLEO missions, the more credible engineering path is a hybrid control architecture, operation at higher altitude, a larger propellant allocation, or an electric-propulsion system with substantially higher effective performance than current small-satellite ion thrusters.

Acknowledgments

References

[1] Japan Aerospace Exploration Agency, "A Variety of Satellites," JAXA Satellite Navigator, n.d. Accessed March 18, 2026.

[2] Japan Aerospace Exploration Agency, "JAXA Terminates the Operation of TSUBAME, a Super Low Altitude Test Satellite (SLATS)," JAXA Press Release, 2019. Press release, October 2, 2019.

[3] NASA Glenn Research Center, "NASA's Evolutionary Xenon Thruster-Commercial (NEXT-C) NEXT Overview," NASA Glenn Fact Sheet, 2021. Fact sheet, November 1, 2021.

[4] Wallace, N., Grubisic, A., Clark, S., Collingwood, C., and Guarducci, F., "Performance Evaluation of the T6 Ion Engine," 46th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, 2010. AIAA 2010-7114.

[5] Krejci, D., Thomas, R., Haag, T., Mier-Hicks, F., and Lozano, P. C., "Emission Characteristics of Passively Fed Electrospray Microthrusters with Propellant Reservoirs," *Journal of Spacecraft and Rockets*, Vol. 54, No. 2, 2017, pp. 447–458.

[6] NASA, "Low-Power Ion Propulsion for Small Spacecraft," Tech. Rep. TM-113111, NASA, 1998.

[7] European Space Agency Advanced Concepts Team, "Dual-stage Gridded Ion Thruster (DS4G)," ESA Advanced Concepts Team project overview, 2006. Project overview, March 3, 2006.

[8] Patterson, M., "Ion Propulsion," 2015.

[9] ESA / Education, "Ion Motors," n.d.

[10] Kumar, M., "Flight Vehicle Attitude Parameterization," 2025.

[11] Kumar, M., "Eulerian Mechanics," 2025.